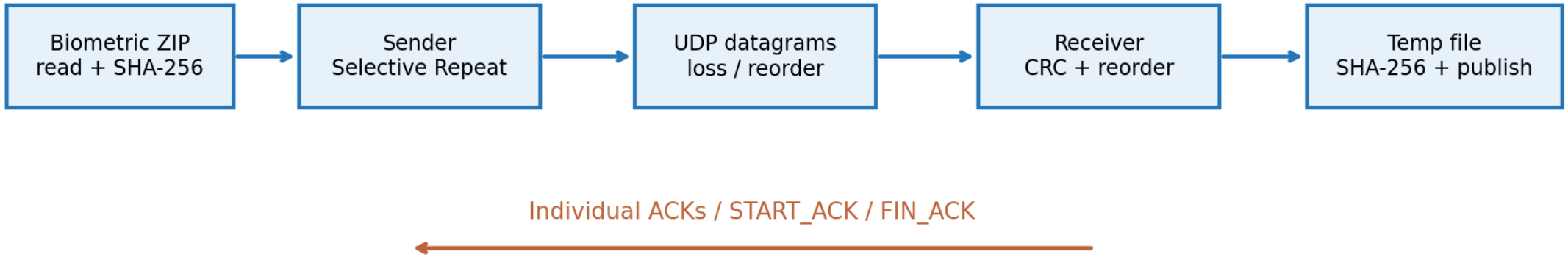
# BioNet | Reliable Biometric Transmission

- Computer Networks: Selective Repeat ARQ over UDP
- Actual biometric ZIP transfer verified on MacBook Air M4
- Python 3.12 • real UDP sockets • CRC32 • SHA-256

# Problem and Objective

- UDP can lose, reorder or duplicate datagrams.
- Transfer captured biometric archive without changing a single byte.
- Implement genuine networking reliability, demonstrate and evaluate.

# System Architecture




Actual sockets on localhost for verified biometric transfer; LAN test not yet demonstrated

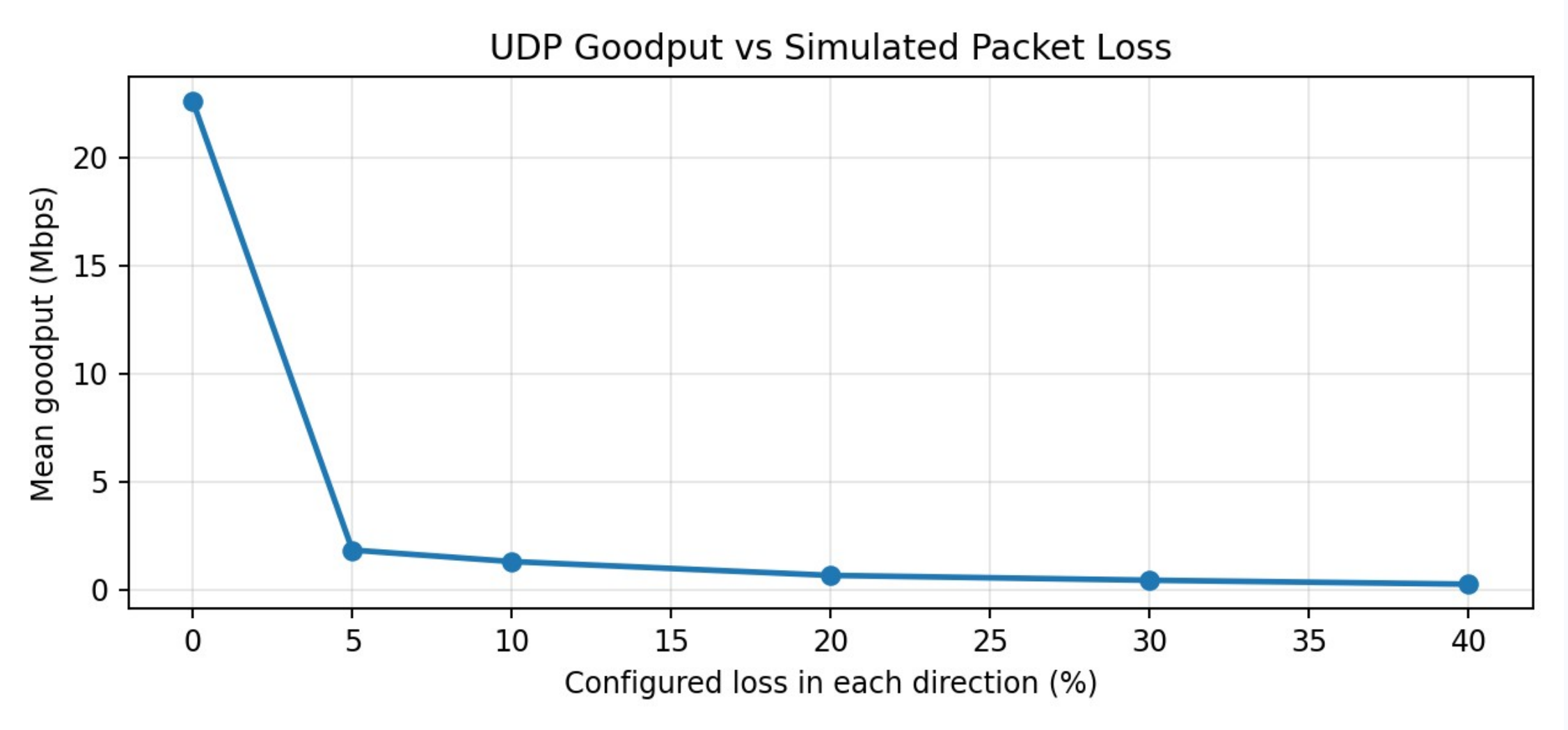
# How Selective Repeat Works

- Sliding window allows multiple outstanding packets.
- Individual ACKs and independent retransmission timers.
- Receiver buffers out-of-order DATA; only missing packets are retried.
- Verified FIN_ACK only after complete SHA-256 match.

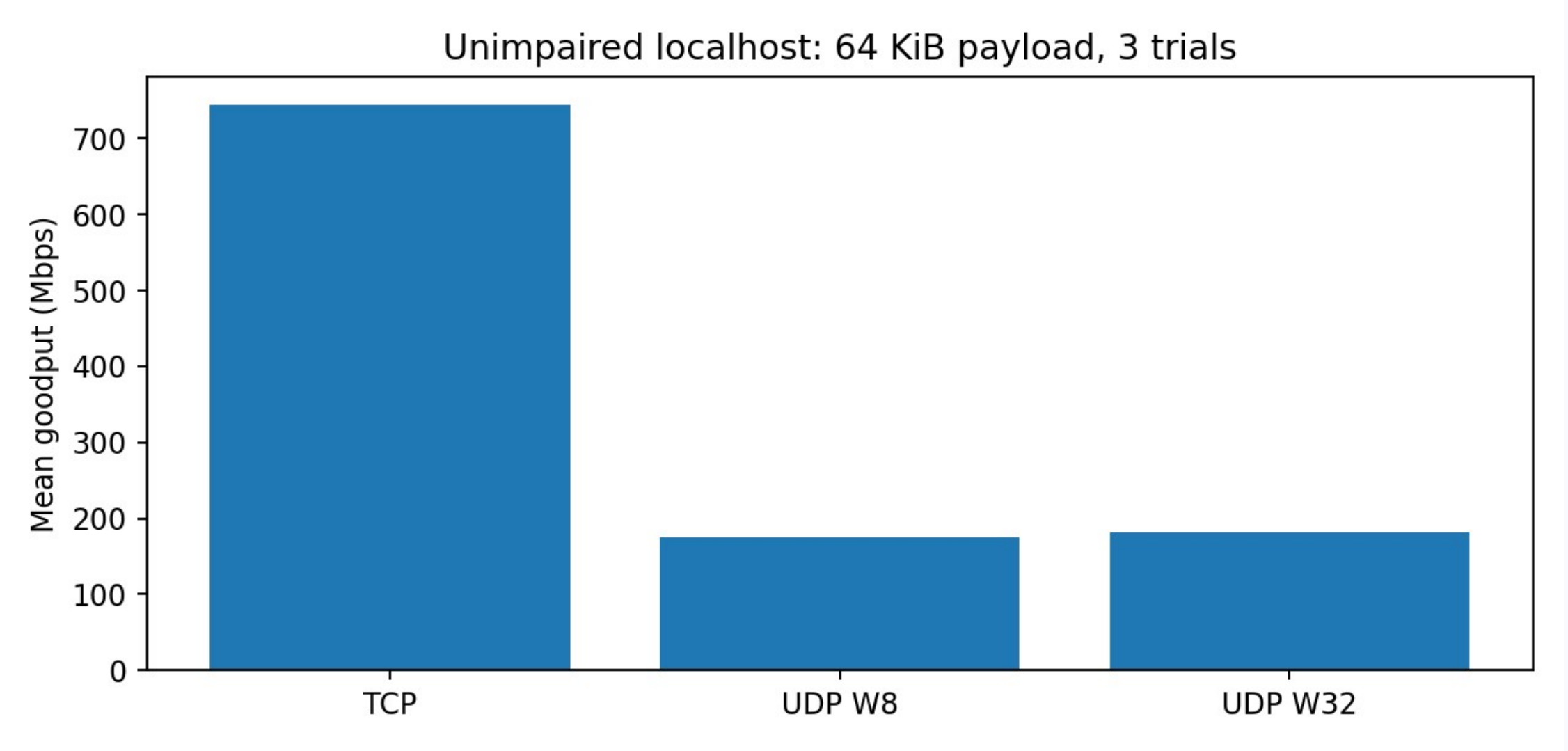
# Real Biometric Transfer — Verified

- 2,774,235,808 bytes • 2,709,215 DATA packets ACKed
- 5% simulated sender loss • 143,000 retransmissions
- 6,453.90 s • final reported goodput 3.439 Mbps
- Sender SHA-256 PASS • receiver SHA-256 PASS • independent cmp PASS

# Measured Performance: Goodput vs Loss



# Measured Performance: TCP vs UDP



# Testing and Limitations

- 181 / 181 automated tests passed on macOS (30.79 s).
- 90 / 90 Phase 3 synthetic benchmark transfers verified.
- Real-data transfer was localhost, not separate-device LAN.
- No encryption, authentication, congestion control or crash-resume.

# Live Demonstration & Conclusion

- Start receiver; send synthetic ZIP with controlled loss.
- Show ACKs, retries, out-of-order buffering and SHA-256 PASS.
- Show actual transfer logs and independent byte-for-byte PASS.
- Reliable reconstruction demonstrated under simulated loss.